

ORIGINAL ARTICLE

Competitive trading in forward and spot markets under yield uncertainty

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Abstract

Many agricultural commodities are traded in both forward and spot markets. This paper studies the interplay of random yield and forward market in a hybrid market with spot and forward transactions. We examine two main questions: (a) How does yield uncertainty (yield risk and yield correlation) affect the equilibrium outcome in this hybrid market? (b) How does the existence of a forward market influence the firms' strategic behaviors and spot price volatility, and how does yield uncertainty mediate the role of the forward market? In our baseline model that considers two firms and no withholding behavior by the firms, it is found that as yield risk increases, firms may sell less in the forward market, and counterintuitively, higher yield risk may benefit firms and make the spot price less volatile. The existence of a forward market leads to greater spot price volatility; that is, a forward market destabilizes spot prices. We identify a mitigating effect of yield variability, but an enhancing effect of yield correlation, on the role of the forward market. Finally, we extend our baseline model to the case with more than two firms and the setting where the firms may withhold some products and demonstrate that some of the key results in the baseline model carry over to these extensions. Nevertheless, the firms' withholding behavior represents a new driving force that changes some results. For example, the decreasing trend of spot price volatility in yield risk disappears in the withholding model.

KEYWORDS

forward market, market equilibrium, price formation, random yield, spot market

1 | INTRODUCTION

Agriculture is an important enabler of the global economy (An et al., 2015). According to the *World Factbook*, agriculture contributed US\$4438 billion, or equivalently a 5.9% share, to the world GDP in 2015. Investments in agriculture have been regarded as a sweet spot for investors. For example, institutional investors such as pension funds see farmland as fertile ground to plow (The Economist, 2015). Moreover, approximately one billion people, or over a third of the available workforce, were employed in the global agricultural sector (International Labor Organization, 2011). Despite the economic and social importance of agriculture, there are significant challenges facing players in this sector. One of the most salient features of agriculture is yield uncertainty. That is, the future output of an agricultural product is

subject to different sources of uncertainty, such as weather and a number of hazards including hurricanes, floods, drought, high-magnitude rainfall, and landslides. Yield risk has been exacerbated by climate change because it causes a higher occurrence of extreme climate events (Vidal, 2013).

There are two aspects of yield uncertainty that warrant special attention. According to an OECD Food, Agriculture and Fisheries Report, a key weather event that affects farming is the rain during the growing season (Kimura & Anton, 2011). The rain varies greatly from one year to the next and has resulted in significant yield risks (i.e., variability of yield). Another characteristic of agricultural yield is its systemic nature. The yield is systemic if farmers face similar yield shocks (e.g., when a disaster hits multiple farmers at the same time). In this paper, we focus on these two measures of yield uncertainty: yield risk (measured by the coefficient of variation) and yield correlation (measured by the correlation coefficient between yields).

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A direct consequence of yield uncertainty is the unpredictable output of an agricultural product and its volatile price. According to Etienne et al. (2017), forward contracting is the principal private market tool utilized by crop farmers in the United States to manage output price risk. In a forward contract, a buyer pays a fixed (forward) price for a fixed amount of products that will be delivered by a seller in the future. In addition, the commodities can be traded in a spot market. The dependence of spot price characteristics (e.g., volatility) on random yield has not received much attention from the literature. There are many existing papers (e.g., Deaton & Laroque, 1992, 1995, 1996; Turnovsky, 1983; Cafiero et al., 2011) that study the commodity price dynamics but without explicitly considering how spot price relies on yield uncertainty. We fill this gap by studying the interplay of random yield and forward market in a hybrid market with spot and forward transactions.

More specifically, our paper attempts to generate operational guidelines for firms' optimal selling and production decisions. For example, the model allows us to examine whether or not the firms should sell more in the forward market when the yield becomes more variable. How do yield uncertainty and forward market jointly affect the firms' production decisions? Will increasing yield risk always make firms worse off? The answer to this question sheds light on whether the firms should make effort to reduce yield uncertainty.

Understanding the implications of yield uncertainty and forward market is not only important for the firms but also provides useful insights to stakeholders who are concerned about spot price volatility. High price volatility is generally perceived as an undesired outcome, which has received much attention from government and academia. For example, during 2007–2008 and 2010–2011, due to worldwide food crises, food price volatility has been brought back to the policy agenda (Bellemare & Lee, 2016). Research has been conducted to examine how to design policies to stabilize the price of agricultural commodities (e.g., Gouel, 2016; Yan, 2016). In this respect, our paper contributes by examining how the interplay of yield uncertainty and forward market affects price volatility in a market with spot and forward transactions. In what situations will reducing yield variability lead to a less volatile spot price? How does yield's systemic nature captured by yield correlation affect spot price volatility? From policymakers' perspective, is establishing a forward market an effective approach to stabilize the spot price?

To address the above research questions, we build a multistage game theoretical model in which firms first determine the amounts of a product to trade in the forward market, then the target production quantities, and, finally, after yield uncertainty is resolved, how much to trade in the spot market. We first characterize the equilibrium for the firms and then conduct a comprehensive sensitivity analysis of the equilibrium outcome with respect to key parameters.

In our baseline model, we study the case with two firms, which is then extended to the case with an arbitrary number of firms in Section 5. In addition, the baseline model

examines situations where firms do not withhold products. This arrangement is appropriate when withholding costs are sufficiently high (e.g., storage and disposal costs are high). In practice, almost all perishable products are difficult to be preserved and stored. Grain crops are suitable for storage but temporary storage can be quite costly due to USDA rules/requirements (where USDA is short for "United States Department of Agriculture") (Kennedy, 2016). Disposal of most products will also incur costs (e.g., landfill costs), and sometimes there are difficulties in product disposal. In England, for instance, according to the Water Industry Act 1991, milk is a kind of "trade effluent" that cannot be discharged into public sewers without consent. Nevertheless, to complement our main analysis, Section 6 relaxes this assumption and considers situations where firms may choose to withhold some products if doing so is economically justifiable.

With the main model setup, we show that the effect of yield risk on firm profits depends on the correlation between firms' yields: if the yields are highly correlated, a higher yield risk always leads to lower firm profits; and if yield correlation is small (e.g., independent yields), each firm's profit first increases and then decreases in the yield risk. This stands in contrast with a result in Deo and Corbett (2009), which show that, in the absence of a forward market, firms' profits always decrease in yield risk when yields are independent. This contrasting result is driven by the fact that the presence of a forward market creates an additional effect of yield risk on firm profits through its impact on *forward positions*. It has been shown that selling in a forward market allows firms to enjoy the first-mover advantage, but in equilibrium firms face a prisoner's dilemma where each firm makes a smaller profit.¹ Increasing yield risk mitigates the first-mover advantage, and thus improves the dilemma situation. Interestingly, we find that an increase in yield risk may benefit firms when yield correlation and yield risk are low.

Our results also show that a higher yield risk may not necessarily translate to higher spot price volatility since price volatility in our model first increases and then decreases in yield risk. We identify that yield risk affects price volatility through both its direct impact and its indirect impact on each firm's target output. The ambiguity of these combined effects arises from the fact that the target output decreases in yield risk. This result suggests that reducing yield uncertainty may backfire in relation to its price volatility implications. In addition, how firms allocate forward sales and spot sales highly depends on yield risk. The intuitive conjecture of higher yield risk leading to higher forward sales does not always hold in our model. When yield correlation is low, firms may engage in a less forward transaction for a higher yield risk. This is because, by spot transaction, a firm can take advantage of a high yield when the other firm's realized yield is low and the spot price is high, which makes forward transaction less attractive to the firm.

Regarding the impact of yield correlation, as the correlation increases, each firm's profit decreases. The intuition is that increasing yield correlation can be regarded as a greater degree of symmetry in terms of ex post yield realizations,

and hence implies more fierce competition on average. We also find that an increase in yield correlation always leads to higher price volatility when there are two firms, but in a market with more firms, a higher yield correlation can result in lower price volatility. To understand this result, we know that spot price volatility increases the volatility of aggregate yield. The volatility of aggregate yield depends on yield correlation directly as well as on its indirect effect on the size of outputs. When yield correlation is higher, the aggregate yield tends to be more volatile due to the lack of risk pooling, which implies a more volatile spot price; on the other hand, a higher yield correlation decreases firms' outputs, which decreases the price volatility. It is found that for two firms the direct effect dominates and a higher yield correlation always results in higher price volatility. But when there is a large number of firms and yield correlation is high, the indirect effect of decreasing output size weighs more, and hence a higher yield correlation may lead to lower price volatility. Moreover, the firms tend to allocate more sales via forward contracts as the correlation increases. To understand it, we note that a forward market brings a first-mover advantage to firms, and this advantage is more prominent in a more competitive market. As discussed earlier, increasing yield correlation implies intensified competition, and hence strengthens the advantage of selling forward for firms. Consequently, firms sell more in the forward market when their yields are more correlated.

Allaz and Vila (1993) have shown, in the absence of yield uncertainty, that the existence of a forward market gets firms trapped in a prisoner's dilemma due to the first-mover advantage. We contribute by showing that increasing yield risk *mitigates*, while increasing correlation *aggravates*, the prisoner's dilemma. We further show that the existence of a forward market leads to higher price volatility.

Finally, although the firms' withholding behavior has an impact on how yield uncertainty affects the equilibrium outcomes, several key insights are robust. For example, as yield risk increases, firms sell less in the forward market, and counterintuitively, higher yield risk may benefit firms; the existence of a forward market leads to higher spot price volatility and lower firm profits. On the other hand, the flexibility of withholding in spot transaction and the lowered effective yield in the spot market for a given target output lead to several distinctive results: the spot price volatility always increases in yield risk, and the effects of yield correlation can be nonmonotone in the withholding model.

The rest of the paper is organized as follows. Section 2 reviews the relevant literature, and Section 3 sets up the model. We analyze the equilibrium for the game and study the role of the forward market in Section 4. To check the robustness of our main results, Sections 5 and 6 examine the case with multiple firms and that with product withholding, respectively. Finally, we conclude and discuss several research extensions in Section 7. The notation table, several extension models, real-world empirical evidence, all the proofs, and some numerical results of the withholding model are provided in the Supporting Information Appendix.

2 | LITERATURE REVIEW

This paper studies the impact of random yield on market equilibrium and spot price volatility. Random yield as a type of supply risk has been extensively studied in operations management; refer to the reviews on lot sizing problems (Yano & Lee, 1995), incentive misalignment issues (Aydin et al., 2011), and supply chain disruption risks (Snyder et al., 2016).

Several recent papers analyze firms' sourcing strategies and supplier competition in the presence of supply risks. Babich et al. (2007) develop a supplier competition model in which a newsvendor sources from multiple unreliable suppliers. Tang and Kouvelis (2011) study firms' sourcing portfolio from reliable and unreliable suppliers when they compete in a downstream market. Hu and Kostamis (2015) consider a manufacturer's sourcing strategy when facing reliable and unreliable suppliers and show that the total order quantity and the order allocation rule are independent decisions. Mu et al. (2021) study a country's food importing strategy when it can negotiate forward and/or ad hoc contracts with outside reliable suppliers. The above papers consider a two-echelon supply chain in which buyers source from unreliable/reliable suppliers. In contrast, we study a horizontal competition setting. Within the horizontal competition paradigm, Deo and Corbett (2009) study a two-stage game with entry and production decisions for a vaccine market. Our paper builds on this paper but differs in several significant ways. First, Deo and Corbett do not consider yield correlation, which is an important characteristic of crop yields. Second, we consider a forward market (again common for commodity products), and one of our focuses is to examine how random yield mediates the role of a forward market. Third, we are interested in price formation in the market with spot and forward transactions and examine the dependence of spot price volatility on yield risks. Due to these differences in model setup, we show some contrasting results that, to our knowledge, are new to the literature.

In this paper, we explicitly consider the dependence of spot price on random yield. In operations management, there is niche literature that considers yield-dependent cost when studying firms' operational decisions. Kazaz (2004) studies an agricultural firm's production planning problem with random yield and random demand where purchasing cost is yield dependent. Here, the purchasing cost refers to the unit cost of purchasing from an outside supplier, and through yield dependence the focal supplier's yield and the outside supplier's price are linked together. Kazaz and Webster (2011) extend Kazaz (2004) by considering both pricing and production planning and show that yield-dependent cost has a profound impact on the firm's optimal decisions. Our paper differs from this literature in several ways: (a) This literature studies a single firm's optimization problem, while we study the strategic interaction between firms; (b) the purchasing cost in these models is an exogenous function of yield, but the price formation is endogenously determined in our model; (c) the present paper focuses on the impact of

random yield on market equilibrium, whereas these papers focus on a firm's production and/or pricing decisions with yield-dependent costs.

This paper is also related to the literature on forward contracts. Several papers in economics study a single firm's decision on forward and spot sales where the distribution of spot price is exogenously given (e.g., Holthausen, 1979; Hey, 1981). With exogenous forward price and spot price, these papers explore how forward price and the distribution of spot price affect the firm's total output. Dong and Liu (2007) consider forward contract negotiation between two risk-averse firms that are price takers in the spot market. On the contrary, this paper explicitly considers spot market price formation with risk-neutral firms and forward buyers. In a competitive setting, several papers study the division between forward market and spot market. Allaz (1992) considers an oligopoly model in which production is conducted *before* demand uncertainty is resolved. Allaz and Vila (1993) study a model with deterministic demand and identify a phenomenon of prisoner's dilemma in which firms prefer to sell forward under competition but become worse off in equilibrium. Popescu and Seshadri (2013) study a similar model to Allaz (1992), but in their paper further production is allowed *after* demand uncertainty is resolved. Our model differs from the above literature by (a) incorporating random yield and (b) considering the production decisions that are made by firms *before* they know the actual yield, which is a salient feature of the agriculture sector.

The effect of forward/futures trading on the spot price volatility has been studied extensively both theoretically and empirically. The results are mixed. Although some empirical works conclude that increasing the financialization of commodity markets stabilizes or at least does not destabilize the spot market (Bohl & Stephan, 2013; Irwin et al., 2009), many others' findings for agricultural commodities, metal, crude oil, and so on are consistent with our result that forward trading destabilizes spot prices (e.g., Gilbert & Morgan, 2010; Nath & Lingareddy, 2008; Slade & Thille, 2006; Yang et al., 2005). Among the theoretical works, Friedman (1953) claims that speculation can stabilize the spot price because, on average, speculators buy when prices are low and sell when they are high. This spirit has been confirmed in subsequent theoretical works (Danthine, 1978; Turnovsky, 1979, 1983; Kawai, 1983). In contrast, some theoretical studies show the destabilizing effect (Hart & Kreps, 1986; Stein, 1987; De Long et al., 1990) by considering traders' inferences and speculators' rational behavior such as buying when the chances of price appreciation are high, and selling when the chances are low. Our work provides a new perspective of the forward market's destabilizing effect by considering producers' selling decisions in the forward market and production decisions in the presence of yield uncertainty.

Finally, our paper is also related to the Cournot competition models with two production periods. Saloner (1987) considers a model where two firms choose initial production quantities in the first period, and after observing the firms' initial production each firm chooses the additional quantity

(which is assumed to be nonnegative) in the second period. A similar model studied by Saloner (1986) allows firms to reduce the first period's production commitment at a cost (i.e., the additional quantity can be negative). This cost endows the firms with a certain degree of commitment to sell what they have produced, thereby restoring the first-mover advantage of production commitment in the first period. Maggi (1996) extends this model by considering demand uncertainty that is realized between the two production periods. Demand uncertainty introduces another driving force so that firms now face a trade-off between flexibility (delaying the investment decision and collecting demand information) and commitment (the first-mover advantage). Similar to the first stage's production commitment in Saloner (1986) and Maggi (1996), forward trading in our model can also be considered as a commitment device for the firms. However, the problem we study is quite different since our focus is on the interplay between yield uncertainty and forward trading as well as their effects on the market equilibrium.

3 | THE MODEL

Our baseline model considers a duopoly setting where there are two key players in the market. Later, in Section 5, we extend the baseline model to settings with more than two firms. The firms engage in quantity competition in the market (a la Cournot model). Each firm i 's yield uncertainty is captured by a random variable Y_i , where $i = 1, 2$. Let y_i be the realization of Y_i . We consider stochastic proportional yields; that is, if firm i 's target quantity is q , then the actual output will be $Y_i q$. Since sources of yield uncertainty (e.g., weather, soil quality, and cropping system) influence the entire crop area, stochastic proportional yield is a reasonable assumption and has been used by many other papers (see e.g., Deo and Corbett, 2009; Kazaz, 2004; Kazaz & Webster, 2011; Tang & Kouvelis, 2011). Our analysis of the model does not require a particular form of yield distribution, but we need to know the mean $E[Y_i] = \mu$ and standard deviation $SD[Y_i] = \sigma$. Define the coefficient of variation for Y_i as $\delta = \sigma/\mu$, so we obtain $\sigma = \mu\delta$. The firms' yields may be correlated and we denote by ρ the correlation coefficient between the yields.² Since the firms growing the same crop and competing in the same market may be located in close proximity, they are likely subject to similar growing conditions (e.g., weather, infestation of pests and diseases, and cropping systems). Thus, it is reasonable to assume that their yields are positively correlated (i.e., $\rho \in [0, 1]$).

Following the previous literature (Deo and Corbett, 2009; Dong et al., 2015), we assume that the inverse demand function is given by $p = a - Q$, where a is the total potential market size and Q is the total output to the market. We assume the market potential a is large enough, that is, $a > b/\mu$, so that the equilibrium production quantity is nonnegative. Firms can sell their products in both forward and spot markets.³ A forward contract is an agreement between the firms and their buyers to deliver a specified

quantity at a later time for a specified price (or in accordance with a specified pricing formula). Depending on terms and conditions, there are different types of forward contracts, including fixed price contract (or flat price), price-to-be-fixed contract, minimum price contract, and so on. We focus on fixed price contract because it is one of the most common types of forward contracts, in which the amount and the price of a commodity are fixed (Kan & Mahajan, 2006).

The model we study is a multistage game. First, firms determine how much to contract in the forward market, f_1 and f_2 . Denote the forward price by p_f , so the forward payment firm i receives is $p_f f_i$, $i = 1, 2$. There are two types of buyers: actual users of the product and speculators. The actual users purchase from the forward market or the spot market. However, speculators only buy from the forward market and then sell the entire quantity to the spot market (Popescu & Seshadri, 2013). Suppose α_i portion of firm i 's forward sales is purchased by the actual users, and $1 - \alpha_i$ portion is purchased by the speculators, where $\alpha_i \in [0, 1]$, $i = 1, 2$.

Second, the firms determine their target outputs, q_1 and q_2 . Specific to agriculture, target outputs refer to an area of land the farmers dedicate to growing crops. The production cost of firm i is bq_i , where b is the unit production cost.⁴

Third, after yield uncertainty is resolved, the forward contracts are executed, and firm i decides the spot transaction amount, s_i , at the spot price p_s . In the baseline model of nonwithholding, firm i sells $s_i^* = y_i q_i - f_i$ in the spot market, where $y_i q_i$ is the realized output. We will relax the nonwithholding assumption in Section 6 by allowing firms to withhold some products if the output yield is very high. A firm's realized output may be less than its forward commitment (i.e., $y_i q_i < f_i$). In this case, s_i^* is negative and the firm purchases the shortage amount, $f_i - y_i q_i$, from the spot market to meet the shortfall of the forward commitment. It is also possible that the firms' aggregate yield, $y_i q_i + y_j q_j$, is lower than their total forward positions, $f_i + f_j$. In this case, they will not be able to buy sufficient amounts from the spot market to fulfill the shortage of the forward contract, $f_k - y_k q_k$, $k = i$ or j . According to a report by Grains Research and Development Corporation (GRDC, an Australian research statutory corporation),⁵ when growers do not produce enough grain and there is no grain available for purchase, they can negotiate a cash settlement with the buyer and the price is usually based on the traded price for grain in the spot market (Grains Research & Development Corporation, 2010). Therefore, in this situation, firm i delivers all it has, $y_i q_i$, and the transaction amount by cash settlement is $s_i^* = -(f_i - y_i q_i)$, which is negative. Firm i 's financial position in the spot market is $-p_s(f_i - y_i q_i) = p_s(y_i q_i - f_i)$, which is the same as the financial position expression when the forward commitment is less than the output yield. Meanwhile, the speculators will sell the products bought in the forward market to the spot market. Note that the market potential in the spot market is $a - (\alpha_1 f_1 + \alpha_2 f_2)$, and the total output to the spot market includes both the speculators' selling quantities and the production firms' sales in the spot market. Hence, the spot

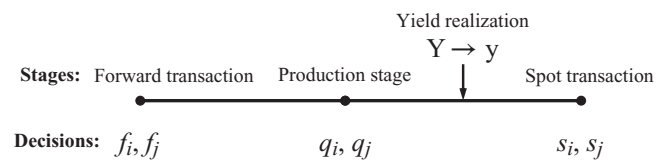


FIGURE 1 The sequence of events, $i, j \in \{1, 2\}, i \neq j$

price is

$$p_s = a - (\alpha_1 f_1 + \alpha_2 f_2) - ((1 - \alpha_1) f_1 + (1 - \alpha_2) f_2) - (s_1 + s_2) \\ = a - \sum_{k=1}^2 (s_k + f_k). \quad (1)$$

Note that the portion of each firm's forward sales purchased by speculators or actual users does not influence the spot price. Intuitively, when a proportion of forward trading is conducted by actual users, the demand of these actual users is satisfied and they will no longer purchase in the spot market. Therefore, compared with the model without actual users, the spot market size will be reduced by the number of the actual users that purchase in the forward market. On the other hand, different from the speculators, actual users will not sell the products purchased in the forward market to the spot market, and hence the total output to the spot market is reduced by the same amount as the actual users. Therefore, the existence of actual users does not influence the spot price.

Finally, the firms and speculators collect payments from the spot sales. The sequence of events is illustrated in Figure 1.

Our model studies a setting in which forward contracts are signed prior to the commencement of production (Turnovsky, 1979). Contracting at an early stage brings several benefits to the agricultural producers (Paul et al., 1976). First, opportunities to sell annual crops several months before planting time give the producers considerable latitude to fix returns at the most opportune time. They can then devote their energies to other matters without worrying about price uncertainties. Second, forward markets can facilitate improved production decisions when used as a guide to determine what and how much to produce. Third, for highly specialized or perishable products, contracting earlier allows the producers to assure outlets for their products.

A final remark is that we consider a full-information setting where production costs and yield distributions are common knowledge to all firms. This is a standard assumption in the literature on forward and spot contracts (Allaz & Vila, 1993; Popescu & Seshadri, 2013). Furthermore, we assume that there are no transaction costs such as costs of carry, product storage, and working capital.

Before analyzing our focal model with a forward market, we first consider a benchmark without a forward market. Comparing the two models allows us to examine the role of the forward market.

3.1 | The benchmark without a forward market

In this model, the firms do not trade in a forward market. A similar model has been studied by Deo and Corbett (2009). Our benchmark extends Deo and Corbett (2009) by considering yield correlation, which is an important point of interest for our study. To differentiate from the focal model with a forward market, we use different notations for target output, spot price, and firms' profits. Let Q_i denote each firm's target output, so the total production cost of firm i is given by bQ_i . Since the firms' withholding behavior is not considered in this benchmark model, the output of firm i is $y_i Q_i$, where y_i denotes the realized yield. Thus, the inverse demand function is given by $P = a - y_1 Q_1 - y_2 Q_2$, where P is the spot price.

First, we consider the firms' best response problems. Given firm j 's target output Q_j , firm i 's objective is to maximize its expected profit by determining an optimal target output. We formulate firm i 's best response problem as follows:

$$\max_{Q_i \geq 0} \mathbb{Z}_i = E[(a - Y_i Q_i - Y_j Q_j) Y_i Q_i - b Q_i]. \quad (2)$$

Each firm maximizes its expected profits in response to the competitor's target output. We summarize the equilibrium for the benchmark model in the following lemma.

Lemma 1. *There exists a unique Nash equilibrium for the game without a forward market.*

(i) *Each firm's target output is given by*

$$Q^* = \frac{a/\mu - b/(\mu^2)}{3 + (2 + \rho)\delta^2}, \quad (3)$$

and each firm's profit is given by

$$\mathbb{Z}^* = \frac{(1 + \delta^2)(a - b/\mu)^2}{(3 + (2 + \rho)\delta^2)^2}. \quad (4)$$

(ii) $\frac{\partial Q^*}{\partial \rho} < 0$, $\frac{\partial \mathbb{Z}^*}{\partial \rho} < 0$, $\frac{\partial Q^*}{\partial \delta} < 0$, and $\frac{\partial \mathbb{Z}^*}{\partial \delta} < 0$.

One observation from the lemma is that each firm's target output and equilibrium profit decrease in yield risk, a result consistent with Deo and Corbett (2009). Another observation is that both the target output and equilibrium profit decrease in yield correlation. The rationale is that increasing yield correlation can be regarded as a greater degree of symmetry in terms of ex post yield realizations and hence implies more fierce competition on average. Lemma 1 also allows us to derive the expected value and the variance of spot price, which are given as follows:

$$E[P^*] = a - \frac{2(a - b/\mu)}{\delta^2(\rho + 2) + 3} \text{ and } \text{Var}[P^*] = \frac{2\delta^2(1 + \rho)(a - b/\mu)^2}{(\delta^2(\rho + 2) + 3)^2}. \quad (5)$$

4 | ANALYSIS FOR THE BASELINE MODEL WITH A FORWARD MARKET

In this section, we consider our focal model where the two firms can trade in a forward market. Given the multistage nature of the game, the solution concept we use is subgame perfect Nash equilibrium. Following backward induction, we start by analyzing the spot and production subgames and then move on to the forward game.

4.1 | Spot and production subgames

In the spot market, the yield uncertainty has been resolved and the forward and production decisions have been made. Firm i determines the transaction amount in the spot market, s_i . As discussed in the model section for the case of no withholding, if the aggregate yield, $y_i q_i + y_j q_j$, is greater than the total forward positions $f_i + f_j$, then $s_i^* = y_i q_i - f_i$ and the spot price in Equation (1) reduces to $a - y_i q_i - y_j q_j$. Otherwise, if the aggregate yield is less than the total forward position and firm i cannot purchase the gap amount $f_i - y_i q_i$ for forward fulfillment, it delivers all it has, $y_i q_i$, and the transaction amount by cash settlement is $s_i^* = -(f_i - y_i q_i)$. In this case, the spot price in Equation (1) is again $a - y_i q_i - y_j q_j$. In all these cases, firm i 's financial position is always $p_s(y_i q_i - f_i)$ in the spot market.

Having established the spot transaction amount, we now consider the firms' production decisions. For the production game, the forward payment $f_i p_f$ is sunk and will not affect the production decision. At the time when the production decision is made, the yield remains uncertain; so firm i 's problem is to maximize its expected profit by choosing an optimal target production quantity. Given firm j 's target output q_j , we formulate firm i 's best response problem as follows:

$$\max_{q_i \geq 0} \Pi_i = E[(a - Y_i q_i - Y_j q_j)(Y_i q_i - f_i)] + p_f f_i - b q_i, \quad (6)$$

where the expectation is taken with respect to Y_i and Y_j . The first and second terms in Equation (6) are firm i 's revenues from spot and forward transactions, respectively.⁶ The total production cost is given in the third term of the objective function. The following lemma characterizes the equilibrium for the production game.

Lemma 2. *Given the forward positions f_1 and f_2 , firm i 's equilibrium target output is given by*

$$q_i(f_i, f_j) = \frac{a/\mu - b/\mu^2}{3 + (2 + \rho)\delta^2} + \frac{2(1 + \delta^2)f_i/\mu - (1 + \rho\delta^2)f_j/\mu}{4(1 + \delta^2)^2 - (1 + \rho\delta^2)^2} \quad (7)$$

for $i = 1, 2, j = 3 - i$.

When $f_1 = f_2 = 0$, we obtain $q_1(0, 0) = q_2(0, 0) = \frac{a/\mu - b/\mu^2}{3 + (2 + \rho)\delta^2}$. Thus, the first term of Equation (7) is the same

as the equilibrium target output in the absence of a forward market (see Equation 3). Lemma 2 shows that each firm i 's equilibrium target quantity is adjusted according to both firms' forward positions. We observe that firm i 's production quantity increases in its own forward position f_i but decreases in the competitor's forward position f_j .

4.2 | Forward game

Next, we consider the forward game. Since there is no information asymmetry among all agents (firms and speculators), their expectations of the future spot price should be identical to the prediction price based on the rational expectations hypothesis proposed by Muth (1961). Furthermore, we consider risk-neutral firms and speculators, and hence the forward price should be an unbiased predictor of the future spot price. To be specific, if the forward price is lower than the expected spot price, the speculators will take advantage of the presented profit opportunity by buying as much as possible in the forward market, which will in turn drive up the forward price due to the high demand. On the other hand, if the forward price is higher than the expected spot price, the speculators will not buy in the forward market, which will reduce the forward price. As a result, the clearing price in the forward market is identical to the expected spot price, that is,

$$p_f = E[a - Y_i q_i(f_i, f_j) - Y_j q_j(f_j, f_i)]. \quad (8)$$

A stream of papers studying commodity pricing with a forward market also adopts this relationship (e.g., Bessembinder & Lemmon, 2002; Routledge et al., 2000; Popescu & Seshadri, 2013).

Given firm j 's forward position f_j , we now examine firm i 's best response in the forward market. Again, in this stage the yields are uncertain, and each firm aims to maximize its expected profit. We formulate firm i 's problem in the forward game as follows:

$$\begin{aligned} \max_{f_i \geq 0} \quad \Pi_i &= E[(a - Y_i q_i - Y_j q_j)(Y_i q_i - f_i)] + p_f f_i - b q_i \\ &= E[(a - Y_i q_i - Y_j q_j)Y_i q_i] - b q_i, \end{aligned} \quad (9)$$

where the second equality is obtained by substituting Equation (8) for p_f . Note that q_i and q_j , as functions of f_i and f_j , are the equilibrium target quantities as shown in Lemma 2. We summarize the equilibrium for the game with a forward market in the following proposition.

Proposition 1 (Equilibrium with the forward market). *There exists a unique subgame perfect Nash equilibrium for the game with the forward market. Each firm's forward quantity is*

$$f^* = \frac{(\delta^2 \rho + 1)^2 (a - b/\mu)}{(-\rho^2 + 2\rho + 4)\delta^4 + 10\delta^2 + 5}, \quad (10)$$

each firm's target output is

$$q^* = \frac{2(\delta^2 + 1)(a/\mu - b/(\mu^2))}{(-\rho^2 + 2\rho + 4)\delta^4 + 10\delta^2 + 5}, \quad (11)$$

and each firm's profit is

$$\Pi^* = \frac{2(\delta^2 + 1)(\delta^4(2 - \rho^2) + 2\delta^2(2 - \rho) + 1)(a - b/\mu)^2}{(\delta^4(-\rho^2 + 2\rho + 4) + 10\delta^2 + 5)^2}. \quad (12)$$

Proposition 1 shows that $f^* > 0$ and $q^* > f^*$, implying that in equilibrium the firms expect to sell in both forward and spot markets. This proposition enables us to analyze the impact of random yield on the equilibrium outcomes in the following subsections.

4.3 | Effect of yield uncertainty

We now examine how yield uncertainty affects the equilibrium outcomes, including equilibrium quantities, profits, and spot price volatility. As discussed earlier, we focus on two measures of yield uncertainty: yield correlation measured by the correlation coefficient ρ , and yield risk measured by the coefficient of variation δ . Hereafter, when studying the effect of δ , we keep the mean μ constant. Therefore, the effect of δ is equivalent to the effect of σ due to the relationship $\delta = \sigma/\mu$.

4.3.1 | Forward quantity and target output

We first examine the effect of yield risk and yield correlation on the forward quantities and the target outputs.

Proposition 2 (Effect of yield uncertainty on forward quantity and target output). *For the game with the forward market, we have*

- (i) *Each firm's target output decreases in both δ and ρ .*
- (ii) *Each firm's forward quantity decreases in δ , but increases in ρ .*

In the absence of a forward market, Deo and Corbett (2009) have shown that firms' target outputs decrease in δ ,⁷ because the marginal revenue $\partial \Pi_i / \partial q_i$ decreases in both target quantity and yield uncertainty. In our model with a forward market, at the time of deciding the target output, the forward quantity decision is sunk. Proposition 2 (i) shows that the presence of a forward market does not change the result of decreasing target outputs in yield uncertainty, since the marginal revenue again decreases in both target quantity and δ .

The marginal revenue $\partial \Pi_i / \partial q_i$ also decreases in ρ , leading to a decreasing target output in ρ . This is because with a

high yield correlation, one firm having a high yield realization means the other firm is likely to also have a high yield. Then the upside of high yield will be dampened by a low spot price and, on average, a higher correlation leads to lower marginal revenue.

To understand the impact of δ and ρ on forward quantities, we investigate the strategic link between forward sales and target outputs. By selling in the forward market (i.e., $f > 0$), a firm is less price sensitive, because the sale in the spot market decreases in f , which leads to a smaller impact of the reduced spot price due to producing a further unit of target output. As a result, the firm selling in the forward market will be more aggressive by targeting a larger output (i.e., q_i increases in f_i). The optimality of a larger target output in the production stage due to selling more in the forward transaction stage is referred to as the first-mover advantage to engage in forward trading.⁸ Therefore, strategically, a firm has an incentive to engage in forward trading in the presence of a forward market (i.e., $f^* > 0$). From Equation (7), we can also examine how δ and ρ influence the responsiveness of target outputs to forward quantities, or equivalently, the significance of first-mover advantage. Note that

$$\frac{\partial q_i}{\partial f_i} = \frac{2(1 + \delta^2)/\mu}{4(1 + \delta^2)^2 - (1 + \rho\delta^2)^2}. \quad (13)$$

It is straightforward to show that $\frac{\partial q_i}{\partial f_i}$ increases in ρ and decreases in δ . That is, a larger yield correlation or a smaller yield risk makes the first-mover advantage of forward trading more significant. This explains the result that the forward quantity increases in yield correlation but decreases in yield risk (Proposition 2(ii)).

4.3.2 | Forward ratio

Define the ratio of forward sales to target outputs as the forward ratio:

$$R = \frac{f^*}{q^*} = \frac{\mu(1 + \rho\delta^2)^2}{2(1 + \delta^2)}. \quad (14)$$

The following proposition summarizes the effect of random yield on the forward ratio.

Proposition 3 (Effect of yield uncertainty on the forward ratio). *For the game with the forward market, we have*

- (i) *If $\rho \geq 1/2$, the forward ratio increases in δ ; if $\rho = 0$, the forward ratio decreases in δ ; if $0 < \rho < 1/2$, the forward ratio first decreases and then increases in δ .*
- (ii) *The forward ratio increases in ρ .*

There is a caveat about the nonmonotone effect of yield risk in Proposition 3(i) for $0 < \rho < 1/2$. For a random vari-

able Y with bounded support $[0,1]$, there is an upper bound M for the coefficient of variation. If the upper bound M is less than the pattern switching threshold of δ (i.e., the value of δ at which the pattern changes from decreasing to increasing, or from increasing to decreasing), the nonmonotone result will vanish and the effect becomes monotone due to the cutoff by the upper bound. Since the upper bound M is different for different distributions, it is infeasible to exhaust all the possibilities for the nonmonotone effect to exist. However, we have analytically and numerically shown that the nonmonotone effect of δ does exist for certain yield distributions.⁹ Therefore, we keep the most general statement such as in Proposition 3. This should be noted hereafter for all results about the nonmonotone effects of δ .

The impact of δ on the forward ratio depends on the yield correlation. When the yield correlation is high ($\rho \geq 1/2$), with a higher yield risk, the ratio of forward sales is always larger. This is because, with the high yield correlation, the upside of a firm's high yield will be dampened by a very low spot price as the other firm's yield is likely to be also high. For a higher yield risk, this dampening effect is stronger and spot transaction becomes less attractive. This suggests that when the yield correlation is high, with higher yield risk, firms tend to "play safe" by selling more in the forward market because doing so guarantees the average spot price (i.e., the forward price). On the other hand, when the firms' yields are independent ($\rho = 0$), with higher yield risk, firms may sell less in the forward market. This is because, with independent yields, spot transaction allows a firm to take advantage of the chance that the spot price may be high when the other firm's realized yield is low, and this benefit is greater for higher yield risk. For $0 < \rho < 1/2$, the forward ratio is not monotone in the yield risk. When the yield risk is small, an increase in yield risk makes spot transaction more attractive, whereas when the yield risk is already large, further increasing yield risk leads firms to "play safe" by selling more in the forward market.

The second part of the proposition shows that a higher yield correlation leads to a higher ratio of forward sales. This result is straightforward because the total output decreases in ρ , while the forward quantity increases in ρ . Allaz and Vila (1993) show in a model without yield uncertainty that the forward ratio depends on market concentration alone. The less concentrated the market is, the higher the ratio. In other words, the more fierce the competition, the higher the ratio. In our model with random yield, as discussed earlier, a higher yield correlation implies more intensified competition and thus leads to a higher forward ratio. For empirical support of this result, it is interesting to contrast crop production with livestock production.¹⁰ Since crop production highly depends on soil type, climate, and the use of irrigation, crop producers tend to congregate in areas with good growing conditions, and thus the correlation of crop production is arguably higher than that of the livestock production. In addition, a 1997 USDA survey shows that marketing contracts are more commonly used in the crop sector than in the livestock sector, thereby aligning with our analytical result that a higher yield correlation leads to a higher ratio of forward sales.

4.3.3 | Profit

We now examine the effect of random yield on each firm's equilibrium profit, which is summarized in the following proposition.

Proposition 4 (Effect of yield uncertainty on profit). *For the game with the forward market, we have*

- (i) *If $\rho \geq 1/2$, then each firm's profit decreases in δ ; but if $\rho < 1/2$, then each firm's profit first increases and then decreases in δ .*
- (ii) *Each firm's profit decreases in ρ .*

Proposition 4(i) shows that firms' profits can be influenced by δ in two directions. This is in contrast with Deo and Corbett (2009), which, assuming no forward market and independent random yield, shows that firms' profits decrease in δ . But in our model, even with $\rho = 0$, firms' profits are nonmonotone in δ . This is because the presence of a forward market delivers another effect of yield uncertainty on firms' profits, which fundamentally changes the result. The effect of δ on profit in Deo and Corbett (2009) is through its impact on target output. With a forward market, the effect of yield uncertainty on profit is also through its impact on the forward quantity. We know that the presence of a forward market allows firms to take the first-mover advantage by engaging in forward transactions. In fact, when both firms have access to a forward market, they are in a prisoner's dilemma, since both trading in the forward market is the equilibrium but it essentially intensifies their competition and makes them worse off. As discussed in Subsection 4.3.1, increasing yield uncertainty decreases the first-mover advantage of engaging in a forward transaction, which reduces firms' incentives to trade in the forward market and thereby dampens their competition. For small ρ and δ values, this force can be so strong that firms' profits increase in δ . Therefore, for small ρ , there is a threshold on yield risk, such that the profits first increase in yield risk when the yield risk is still below the threshold and then decrease in yield risk when above the threshold.¹¹ This suggests that a moderate level of yield risk can actually benefit the firms by reducing their competition intensity. Thus, it is not always worthwhile making an effort to reduce yield risk.

Conversely, the presence of a forward market does not change the negative impact of increasing ρ on firms' profits. This is because the first-mover advantage is amplified by an increase in yield correlation, which leads firms to be more aggressive by selling more in the forward market and thereby makes them even worse off. That is, a larger yield correlation puts firms in a more severe prisoner's dilemma. Therefore, firms' profits always decrease in ρ , no matter whether the forward market is present or absent.

4.3.4 | Spot price

Based on Proposition 1, we are able to characterize the spot price by calculating its expected value and variance, which are given as follows:

$$E[p^*] = a - \frac{4(a - b/\mu)(\delta^2 + 1)}{\delta^4(-\rho^2 + 2\rho + 4) + 10\delta^2 + 5} \text{ and}$$

$$\text{Var}[p^*] = \frac{8\delta^2(\delta^2 + 1)^2(\rho + 1)(a - b/\mu)^2}{(\delta^4(-\rho^2 + 2\rho + 4) + 10\delta^2 + 5)^2}. \quad (15)$$

The following proposition investigates the dependence of spot price on yield risk and correlation.

Proposition 5 (Effect of yield uncertainty on spot price). *For the game with the forward market, we have*

- (i) *The expected spot price increases in both ρ and δ .*
- (ii) *The variance of spot price increases in ρ ; while it first increases and then decreases in δ .*

The above proposition reveals that the expected spot price increases in both yield risk and yield correlation. This result is easy to understand. Proposition 2 shows that each firm's target output decreases in both yield risk and yield correlation. We also know that $E[p^*] = a - \mu q_i^* - \mu q_j^*$, implying that the expected spot price decreases in each firm's target output. Therefore, as yield risk or yield correlation increases, the average spot price increases.

Proposition 5(ii) first shows that the spot price variance increases in the yield correlation. The intuition is that spot price volatility in our model increases in the volatility of aggregate yield. When the yield correlation is high, the aggregate output becomes more volatile due to the lack of a risk-pooling effect. Thus, the spot price becomes more volatile when firms' yields are more correlated. This result is aligned with the finding of a USDA report that "crop prices are more volatile than livestock prices" (Harwood et al., 1999) because, as discussed earlier, crop production tends to have a higher correlation than livestock production.

Proposition 5(ii) also shows that spot price volatility first increases and then decreases in yield risk. The proof of the proposition shows that $\text{Var}[p^*] = (2 + 2\rho)\mu^2(q^*)^2\delta^2$. Note that q^* is a function of δ , and thus the effect of yield risk on spot price volatility depends on (a) its impact on each firm's target outputs and (b) its direct impact. More formally, we have $\partial \text{Var}[p^*]/\partial \delta = (2 + 2\rho)\mu^2[2q^*\delta^2(\partial q^*/\partial \delta) + 2(q^*)^2\delta]$. Proposition 2 has shown that the target output decreases in yield risk, that is, $\partial q^*/\partial \delta < 0$. The combined effect of the above two forces reveals that yield risk has a nonmonotone effect on spot price variance. When the yield risk is still small, the target output is high. This makes the direct effect

of increasing yield risk on spot price volatility dominate the indirect effect of decreasing target outputs, and hence the volatility increases in the yield risk. But when the yield risk is large, the target output is small. In this case, the indirect effect on the target outputs dominates, and the volatility decreases in the yield risk. It is counterintuitive that a higher yield risk leads to a lower spot price volatility. However, this result is supported by the following practical evidence. Harwood et al. (1999) show that corn yield variability tends to be the lowest in the central U.S. Corn Belt, where soils are deep and rainfall is dependable, but on the border of the Corn Belt, corn acreage tends to be fairly low and yield variability is generally higher. By analyzing the U.S. state-level corn price data (1985–1994), we find that the coefficient of variation of corn price is relatively low in the states located on the border of the Corn Belt (e.g., Minnesota, South Dakota, and Wisconsin). That is, consistent with our result, in areas with high yield risk, farmers choose to limit the acreage planted with corn because of the unstable condition for corn production, and, as a result, the spot price variability of corn is relatively lower than in other areas.

In the proof of Proposition 5, we have confirmed that the coefficient of variation for the spot price, $cv[p^*] = \frac{SD[p^*]}{E[p^*]}$, increases in the yield correlation, but first increases and then decreases in the yield risk. This demonstrates that the impact of yield risk on spot price volatility (measured by both variance and coefficient of variation) is not monotone.

4.4 | Role of a forward market

In this subsection, we study the role of a forward market and how it is mediated by yield uncertainty. Comparing the cases with and without a forward market, we find the following results.

Proposition 6 (Role of a forward market). *We have*

- (i) $q^* > Q^*$, $\frac{\partial(q^*-Q^*)}{\partial\rho} > 0$, and $\frac{\partial(q^*-Q^*)}{\partial\delta} < 0$;
- (ii) $\Pi^* < Z^*$, $\frac{\partial(\Pi^*-Z^*)}{\partial\rho} > 0$, and $\frac{\partial(\Pi^*-Z^*)}{\partial\delta} < 0$;
- (iii) $E[p^*] < E[P^*]$, $\frac{\partial(E[p^*]-E[P^*])}{\partial\rho} > 0$, and $\frac{\partial(E[p^*]-E[P^*])}{\partial\delta} < 0$;
- (iv) $\text{Var}[p^*] > \text{Var}[P^*]$, $\frac{\partial(\text{Var}[p^*]-\text{Var}[P^*])}{\partial\rho} > 0$, and $\frac{\partial(\text{Var}[p^*]-\text{Var}[P^*])}{\partial\delta} < 0$.

With a forward market, the total output is larger and the expected spot price is smaller. However, each firm becomes worse off in the presence of a forward market. A similar result has been found by Allaz and Vila (1993), who study a deterministic model where there is no yield uncertainty. As discussed earlier, firms enjoy a first-mover advantage by selling forward. As a result, both firms increase their target outputs, which leads to a lower spot price and erodes each firm's profitability.

New to the literature, this proposition shows that, in the presence of yield uncertainty, the forward market makes the spot price more volatile. We know that the variance of spot price increases in each firm's target outputs. Because the target outputs are higher when there is a forward market, the spot price becomes more volatile. In the presence of a forward market, since the expected spot price is smaller while the variance of spot price is higher, it is straightforward that the coefficient of variation for the spot price is also higher, that is, $\frac{SD[p^*]}{E[p^*]} > \frac{SD[P^*]}{E[P^*]}$. This result can explain the empirical findings that forward/futures trading destabilizes spot prices of some commodities (e.g., Nath & Lingareddy, 2008; Slade & Thille, 2006; Yang et al., 2005; Gilbert & Morgan, 2010). In addition, some research reports also document the destabilizing effect of the forward/futures market. A report of the European Parliament states that financial speculation in commodity derivatives markets has been seen as one of the drivers of the peaks in agricultural prices in 2007–2008 and 2010–2011 (Azevedo et al., 2013). Speculative fund activities in futures markets have led to more money and higher volatility in the agricultural market (Agricultural and Food Policy Center, 2008). The derived result that a forward market destabilizes spot price may have some policy implications. For instance, the Central Bank of Sri Lanka established a forward market (i.e., Forward Sales Contracts) in order to resolve the major issues in the agricultural sector such as high price fluctuations. Our results suggest that policymakers' strategy of establishing a forward market may have unintended consequences.

Proposition 6 further shows that, as the yield risk increases, the impact of a forward market on all aspects of the equilibrium outcome (e.g., each firm's equilibrium profit and spot price volatility) is mitigated. In contrast, increasing yield correlation enhances the impact of the forward market on the equilibrium results. That is, a larger yield correlation puts firms in a more severe situation of prisoner's dilemma, but a higher yield risk mitigates the dilemma issue.

5 | THE CASE WITH MULTIPLE FIRMS

So far we have considered the case with two firms. This section extends the model to the case with $n > 2$ firms. As before, we focus on ex ante symmetric firms where each firm's yield follows the same distribution.

The set of suppliers is denoted by $N = \{1, \dots, n\}$. For tractability, we assume the correlation between any two firms' yields is ρ , so $E[Y_i Y_j] = \mu^2 + \rho\sigma^2$ for any $i, j \in N$ and $i \neq j$. As mentioned in Section 3, we consider nonnegative yield correlations (i.e., $\rho \in [0, 1]$). Following backward induction, we start by analyzing the firms' decisions in the spot market. Similar to the two-firm case, we consider no withholding: The firms output all the remaining products (if any) to the spot market after meeting their forward obligations, and in case of supply shortage, they may purchase from the spot market or negotiate a cash settlement to meet the shortfall of forward commitments. Thus, for any given realized

yields $\{y_i : i \in N\}$, firm i 's optimal spot transaction amount is $s_i^* = y_i q_i - f_i$, and, correspondingly, the spot price is given by $p_s = a - \sum_{j=1}^n y_j q_j$. Again we assume $a > b/\mu$ so that the equilibrium target output is always positive.

We next examine the production game. Let p_f be the forward price, and $\mathbf{f} = \{f_j, j \in N\}$ be the set of forward positions. Given the other firms' target outputs $\mathbf{q}_{-i} = \{q_j, j \in N, j \neq i\}$, each firm i aims to maximize its expected profit by choosing an optimal target output. Firm i 's problem is formulated as follows:

$$\max_{q_i \geq 0} \Pi_i = \mathbb{E} \left[\left(a - \sum_{j=1}^n Y_j q_j \right) (Y_i q_i - f_i) \right] + f_i p_f - b q_i. \quad (16)$$

Jointly solving the above optimization problem for all firms, we obtain the equilibrium target outputs in the following lemma.

Lemma 3. *Given the forward positions $\mathbf{f}$, firm i 's equilibrium target output is given by*

$$q_i(\mathbf{f}) = \frac{a/\mu - b/\mu^2}{(n+1) + (2 + (n-1)\rho)\delta^2} + \frac{(n + (2 + (n-2)\rho)\delta^2)f_i/\mu - (1 + \rho\delta^2) \sum_{j \neq i} f_j/\mu}{(1 + (2 - \rho)\delta^2)((n+1) + (2 + (n-1)\rho)\delta^2)}. \quad (17)$$

The above lemma characterizes the equilibrium for the production game given fixed forward quantities $\mathbf{f}$. The results allow us to examine how a firm's forward quantity affects its equilibrium target output. Taking the derivative of $q_i(\mathbf{f})$ with respect to f_i , we obtain

$$\frac{\partial q_i(\mathbf{f})}{\partial f_i} = \frac{\delta^2((n-2)\rho + 2) + n}{\mu(\delta^2(2 - \rho) + 1)(\delta^2((n-1)\rho + 2) + n + 1)}. \quad (18)$$

This measure can be regarded as a proxy for the significance of first-mover advantage, as discussed after Proposition 2. We will use this measure to explain the sensitivity analysis results.

We now look at the forward game for the n firms. Again we consider a frictionless forward market in which the forward price is equal to the expected spot price. That is, $p_f = \mathbb{E}[a - \sum_{j=1}^n Y_j q_j(\mathbf{f})]$. Substituting this equation into Π_i , we obtain

$$\begin{aligned} \Pi_i &= \mathbb{E} \left[\left(a - \sum_{j=1}^n Y_j q_j(\mathbf{f}) \right) (Y_i q_i(\mathbf{f}) - f_i) \right] + f_i \mathbb{E} \left[a - \sum_{j=1}^n Y_j q_j(\mathbf{f}) \right] - b q_i(\mathbf{f}) \\ &= \mathbb{E} \left[\left(a - \sum_{j=1}^n Y_j q_j(\mathbf{f}) \right) Y_i q_i(\mathbf{f}) \right] - b q_i(\mathbf{f}), \end{aligned} \quad (19)$$

where $q_i(\mathbf{f})$ is the equilibrium production quantity of firm i as shown in Equation (17). Each firm maximizes its expected profit by choosing an optimal forward position. The following proposition characterizes the equilibrium for the firms.

Proposition 7 (Equilibrium with multiple firms). *When there are n firms, there exists a unique subgame perfect Nash equilibrium. Each firm's forward quantity is given by*

$$f^* = \frac{(n-1)(a - b/\mu)(1 + \rho\delta^2)^2}{\delta^4((n^2 - 4n + 3)\rho^2 + (4n - 6)\rho + 4) + 2\delta^2(n^2\rho + 2n(1 - \rho) + 1) + (n^2 + 1)}, \quad (20)$$

each firm's target output is given by

$$q^* = \frac{(a/\mu - b/\mu^2)(2(1 - \rho)\delta^2 + n(1 + \rho\delta^2))}{\delta^4((n^2 - 4n + 3)\rho^2 + (4n - 6)\rho + 4) + 2\delta^2(n^2\rho + 2n(1 - \rho) + 1) + (n^2 + 1)}, \quad (21)$$

and each firm's profit is given by $\Pi^ = (a\mu - (n-1)(\mu^2 + \rho\sigma^2)q^* - (\mu^2 + \sigma^2)q^*)q^* - bq^*$.*

Next, we study how yield uncertainty and market concentration (measured by n) affect the two important equilibrium outcomes: forward ratio and spot price volatility. We will demonstrate that the key results for the two-firm case carry over to the n -firm case and also show some new results regarding the impact of n .

Forward ratio

We begin by looking at the forward ratio. Based on the results in the above proposition, we calculate the forward ratio as follows:

$$R = \frac{f^*}{q^*} = \frac{\mu(n-1)(\delta^2\rho + 1)^2}{2\delta^2(1 - \rho) + \delta^2 n\rho + n}. \quad (22)$$

The following proposition summarizes the comparative statics results with respect to the number of firms, yield risk, and yield correlation.

Proposition 8 (Effects of yield uncertainty and number of firms on the forward ratio). *For the game with n firms, we have*

- (i) *The forward ratio increases in n .*
- (ii) *The results in Proposition 3 regarding the effect of δ and ρ on the forward ratio under the two-firm case carry over to the n -firm case.*

Part (i) of Proposition 8 shows that, as the number of firms increases, the forward ratio also increases. That is, more intense competition leads firms to sell more in the forward market, and hence results in a higher forward ratio. This result relates to the fact that the first-mover advantage, measured

by $\partial q_i(\mathbf{f})/\partial f_i$ in Equation (18), increases in n . This implies that more firms in the market make the first-mover advantage stronger, explaining why the forward ratio increases in n .

Part (ii) of the above proposition demonstrates that the results about the effects of yield risk and yield correlation on the forward ratio are robust to the n -firm case. In particular, regarding the effect of yield risk on the forward ratio, we find that if $\rho > 2/(2+n)$, the forward ratio increases in δ ; but if $\rho < 2/(2+n)$, the forward ratio decreases in δ for $\delta < \sqrt{2-(n+2)\rho}/\sqrt{(n-2)\rho^2+2\rho}$ and increases in δ for $\delta > \sqrt{2-(n+2)\rho}/\sqrt{(n-2)\rho^2+2\rho}$. It is straightforward to observe that, $\rho > 2/(2+n)$ and $\delta > \sqrt{2-(n+2)\rho}/\sqrt{(n-2)\rho^2+2\rho}$ are easier to be satisfied for a larger n . This implies that when there are more firms in the market, it is more likely for $\partial R/\partial \delta > 0$ to hold. That is, if the market competition is intense, the firms tend to “play safe” by selling more in the forward market as the yield risk increases.

Spot price

We now consider the properties of the equilibrium spot price. Note that $E[p] = E[a - \sum_{j=1}^n Y_j q_j] = a - \mu \sum_{j=1}^n q_j$, and $\text{Var}[a - \sum_{j=1}^n Y_j q_j] = \sum_{j=1}^n (q_j)^2 \sigma^2 + \sum_{j=1}^n (\sum_{k \neq j} q_k \rho \sigma^2)$.

Since the sensitivity analysis of the spot price is quite messy for an arbitrary n , we focus on the asymptotic behavior as n approaches infinity. We obtain

$$\lim_{n \rightarrow \infty} E[p^*] = \frac{a\delta^2\rho + b/\mu}{\delta^2\rho + 1} \text{ and } \lim_{n \rightarrow \infty} \text{Var}[p^*] = \frac{\rho\delta^2(a - b/\mu)^2}{(\delta^2\rho + 1)^2}. \quad (23)$$

Again, we can show that a higher yield risk or yield correlation leads to a higher expected spot price, which is consistent with the two-firm results as shown in part (i) of Proposition 5. However, the impact of yield correlation on the spot price variance for the n -firm case turns out to be different. We summarize this result in Proposition 9.

Proposition 9 (Effect of yield uncertainty on spot price). *When the number of firms goes to infinity, we have*

- (i) *The spot price variance increases in ρ for $\rho < 1/\delta^2$ and decreases in ρ for $\rho > 1/\delta^2$.*
- (ii) *The spot price variance first increases and then decreases in δ , consistent with its counterpart in Proposition 5.*

Recall from Proposition 5 that under the two-firm setting, the variance of spot price increases in ρ . Proposition 9 reveals that when the number of firms goes to infinity, the spot price variance can be lower for a larger yield correlation under the condition $\rho > 1/\delta^2$. This contrasting result is driven by the effect of yield correlation on the size of outputs. We know that spot price volatility increases in the volatility of aggregate yield. The volatility of aggregate yield depends on yield correlation directly as well as on its indirect effect on the size of outputs. The direct effect of a higher yield correlation is

a more volatile spot price due to the lack of risk pooling. However, the indirect effect of correlation leads to smaller outputs of firms, thereby reducing the price volatility. For a small number of firms, such as two firms, the indirect effect is weak and is always dominated by the direct effect, and hence a higher yield correlation always results in higher price volatility. But when the number of firms is large and yield correlation is high, the indirect effect of decreasing output size is more significant, and hence a higher yield correlation leads to lower price volatility. The proposition also shows that the spot price variance in the limit first increases and then decreases as the yield risk increases, a result consistent with the two-firm setting as shown in Proposition 5.

To check the robustness of the results in Proposition 9 using variance as the volatility measure, in the proof of this proposition we also study the sensitivity of the coefficient of variation with respect to the yield risk and correlation. We find that the sensitivity results using the coefficient of variation to measure the volatility have the same pattern as those using variance as the measure.

6 | EFFECT OF FIRMS' WITHHOLDING BEHAVIOR

This section considers the case where firms may not output all remaining products to the spot market after meeting the forward obligation. We begin by presenting the formulation of the general problem and then, for tractability purposes, analyze a special case where the firms' yields follow a joint two-point distribution.

6.1 | Model formulation

6.1.1 | Spot game

In the spot subgame, the firms' forward payments and production costs are sunk. Suppose the firms' target outputs are q_i and q_j , and the forward positions are f_i and f_j . Then firm i 's optimization problem in the spot market is formulated as follows:

$$\begin{aligned} \max_{s_i} \quad & \Pi_i = (a - s_i - f_i - s_j - f_j)s_i + p_f f_i - b q_i, \\ \text{s.t.} \quad & s_i \leq y_i q_i - f_i. \end{aligned} \quad (24)$$

The first and second terms of the objective function represent firm i 's revenue from the spot and forward markets, respectively. The third term is the total production cost. The constraint states that a firm's total sales via forward and spot transactions cannot exceed the realized total yield.

It is easy to derive firm i 's best response spot transaction quantity as follows:

$$s_i^*(s_j) = \min \left\{ \frac{a - s_j - f_i - f_j}{2}, y_i q_i - f_i \right\}, \text{ for } i = 1, 2, j = 3 - i. \quad (25)$$

Note that a firm may not output all the remaining products to the spot market after fulfilling the forward contract. This occurs when its realized yield is high. Further, unlike the nonwithholding setting, the firms' optimal spot transaction quantities are interdependent in this extension.

Jointly solving the best response equations in the above, we characterize the equilibrium for the spot market subgame. Depending on the realized yields (y_i, y_j) , there are four cases of the equilibrium spot transaction quantities:

$$(s_i^*, s_j^*) = \begin{cases} \left(\frac{a - f_i - f_j}{3}, \frac{a - f_i - f_j}{3} \right), & \text{if } (y_i, y_j) \in \Omega_1, \\ \left(\frac{a - y_j q_j - f_i}{2}, y_j q_j - f_j \right), & \text{if } (y_i, y_j) \in \Omega_2, \\ \left(y_i q_i - f_i, \frac{a - y_i q_i - f_j}{2} \right), & \text{if } (y_i, y_j) \in \Omega_3, \\ (y_i q_i - f_i, y_j q_j - f_j), & \text{if } (y_i, y_j) \in \Omega_4, \end{cases} \quad (26)$$

where

$$\begin{aligned} \Omega_1 &= \{y_i, y_j : 3y_i q_i \geq a + 2f_i - f_j, 3y_j q_j \geq a + 2f_j - f_i\}, \\ \Omega_2 &= \{y_i, y_j : 2y_i q_i + y_j q_j \geq a + f_i, 3y_j q_j \leq a + 2f_j - f_i\}, \\ \Omega_3 &= \{y_i, y_j : 3y_i q_i \leq a + 2f_i - f_j, y_i q_i + 2y_j q_j \geq a + f_j\}, \\ \Omega_4 &= \{y_i, y_j : 2y_i q_i + y_j q_j \leq a + f_i, y_i q_i + 2y_j q_j \leq a + f_j\}. \end{aligned} \quad (27)$$

The realized yield space (y_i, y_j) is separated into four regions and is illustrated in Figure 2. As can be seen, both firms withhold some products when the realized yields lie in region Ω_1 , firm i withholds when the yields lie in region Ω_2 , and firm j withholds when the yields are in Ω_3 .

Note from Figure 2 that when $\frac{a+2f_i-f_j}{3q_i} > 1$ for $i = 1, 2, j = 3 - i$, the yield realizations always lie in Ω_4 , so withholding never happens. This degenerated scenario is covered in our baseline model.

6.1.2 | Production game

We now consider the firms' production decisions. For convenience, we denote by $s_i^*|_{\Omega_k}$ the equilibrium spot transaction quantity of firm i associated with region Ω_k . In the production subgame, the firms' forward payments are sunk. Recall that the forward positions are f_i and f_j . We formulate firm i 's best response problem for the production game as follows:

$$\begin{aligned} \max_{q_i \geq 0} \quad \Pi_i &= E[(a - s_i^* - f_i - s_j^* - f_j)s_i^*] + p_f f_i - b q_i \\ &= \sum_{k=1,2,3,4} \int_{\Omega_k} (a - s_i^*|_{\Omega_k} - f_i - s_j^*|_{\Omega_k} - f_j) s_i^*|_{\Omega_k} dG(y_i, y_j) \\ &\quad + p_f f_i - b q_i, \end{aligned} \quad (28)$$

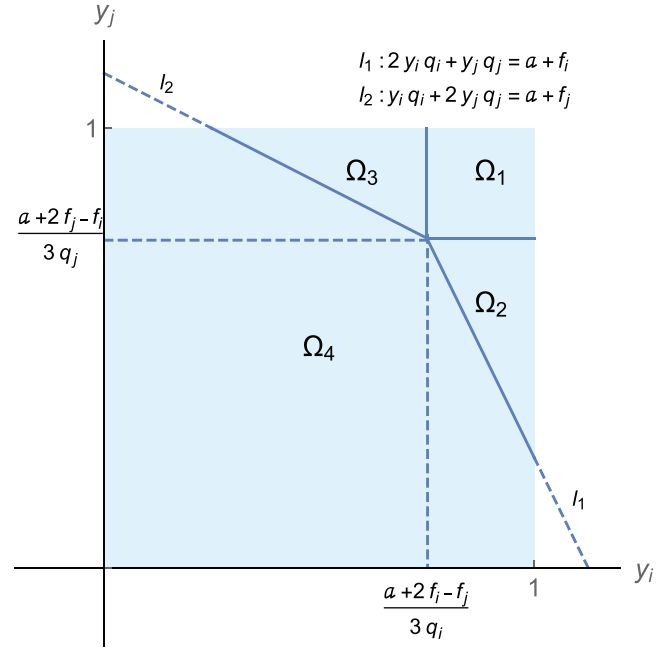


FIGURE 2 Illustration of the four regions in (y_i, y_j) space for the spot game [Color figure can be viewed at wileyonlinelibrary.com]

where $G(y_i, y_j)$ is the joint cumulative distribution function of Y_i and Y_j . Note that $s_i^*|_{\Omega_k}$ and $s_j^*|_{\Omega_k}$ are functions of q_i and q_j for $k = 1, \dots, 4$. The next task is to characterize the equilibrium for the production subgame by jointly solving the two firms' best response problems. Since the equilibrium spot quantities take several different forms and the objective function involves integration over multiple regions, this task presents a significant challenge for a general probability distribution. To move the analysis forward, in Subsection 6.2 we will consider a joint two-point distribution. Let the equilibrium target outputs be $q_i^*(f_i, f_j)$, $i = 1, 2, j = 3 - i$.

6.1.3 | Forward game

Next, we consider the forward game played by the firms. When the firms can withhold some products in the spot market, firm i 's best response problem in the forward game is formulated as follows:

$$\max_{f_i \geq 0} \quad \Pi_i = \sum_{k=1,2,3,4} \int_{\Omega_k} (a - s_i^*|_{\Omega_k} - f_i - s_j^*|_{\Omega_k} - f_j) (s_i^*|_{\Omega_k} + f_i) dG(y_i, y_j) - b q_i^*(f_i, f_j), \quad (29)$$

where $s_i^*|_{\Omega_k}$ and $s_j^*|_{\Omega_k}$ are the equilibrium spot transactions evaluated at $q_i = q_i^*(f_i, f_j)$ and $q_j = q_j^*(f_j, f_i)$, and hence both are functions of f_i and f_j . Note that Equation (29) has used the assumption that the forward price is an unbiased predictor of the spot price. The next step is to jointly solve the above optimization problems for both firms and

TABLE 1 Patterns of firms' withholding behaviors in the model with a joint two-point distribution

Patterns	Conditions on the four possible yield realizations
(I) Firms withhold some outputs if and only if they both realize a high yield	$(y_i^l, y_j^l), (y_i^l, y_j^h)$ and $(y_i^h, y_j^l) \in \Omega_4, (y_i^h, y_j^h) \in \Omega_1$
(II) A firm withholds some outputs as long as it realizes a high yield	$(y_i^l, y_j^l) \in \Omega_4, (y_i^l, y_j^h) \in \Omega_3, (y_i^h, y_j^l) \in \Omega_2, (y_i^h, y_j^h) \in \Omega_1$
(III) Firms always withhold some outputs	$(y_i^l, y_j^l), (y_i^l, y_j^h), (y_i^h, y_j^l)$, and $(y_i^h, y_j^h) \in \Omega_1$
(IV) Firms never withhold outputs	$(y_i^l, y_j^l), (y_i^l, y_j^h), (y_i^h, y_j^l)$, and $(y_i^h, y_j^h) \in \Omega_4$

obtain each firm's equilibrium forward quantity and profit. Again, this analysis is challenging for a general yield distribution.

6.2 | The case with a two-point yield distribution

For tractability, we consider the following two-point yield distribution: $Pr(Y_i = y_i^h) = Pr(Y_i = y_i^l) = 1/2$, where $y_i^h = 1/2 + \sigma$ and $y_i^l = 1/2 - \sigma$ for $i = 1, 2$. We assume $\sigma \in (0, 1/2)$ to ensure that the yield is between zero and one. With this distribution, the mean yield is $\mu = 1/2$ and the standard deviation of the yield is equal to σ . Thus, the coefficient of variation is given by $\delta = \sigma/\mu = 2\sigma$ ($0 < \delta < 1$). The two random variables, Y_i and Y_j , may be correlated, and this is captured by their joint probabilities: $Pr(Y_i = y_i^l, Y_j = y_j^l) = Pr(Y_i = y_i^h, Y_j = y_j^h) = \tau$ and $Pr(Y_i = y_i^l, Y_j = y_j^h) = Pr(Y_i = y_i^h, Y_j = y_j^l) = \frac{1}{2} - \tau$, where we assume $1/4 \leq \tau \leq 1/2$ to ensure a nonnegative correlation between the yields. Babich et al. (2007) and Tang and Kouvelis (2011) use a similar joint distribution to analyze the impact of supply correlation. It follows from the joint distribution that the correlation coefficient between Y_i and Y_j , denoted by ρ , equals $4\tau - 1$. In particular, $\tau = 1/2$ represents the case with perfect correlation and $\tau = 1/4$ represents the case with independent yields.

For this two-point distribution, we use (y_i^l, y_j^l) , (y_i^l, y_j^h) , (y_i^h, y_j^l) , and (y_i^h, y_j^h) to denote the four possible scenarios of yield realizations. As expected, the values of y_i^l and y_i^h play an important role in determining the firms' withholding decisions. Depending on parameter settings, there are various patterns for the firms' withholding decisions, summarized in Table 1.

Note first that Pattern IV corresponds to our baseline model. Further, Pattern III will never emerge in equilibrium because the firms can always save costs by simply cutting their target outputs should they need to withhold products under all scenarios. In the subsequent analysis, we will focus on Pattern II, where firm i withholds some outputs as long as the realization of Y_i is y_i^h , $i = 1, 2$. We have also checked that the sensitivity analysis results for Pattern I have similar structures to those of Pattern II. Note from Figure 2 that the condition of Pattern II to guarantee $(y_i^l, y_j^l) \in \Omega_4$, $(y_i^l, y_j^h) \in \Omega_3$, $(y_i^h, y_j^l) \in \Omega_2$, and $(y_i^h, y_j^h) \in \Omega_1$ is

given by

$$\frac{a + f_j - 2y_j^h q_j}{q_i} < y_i^l < \frac{a + 2f_i - f_j}{3q_i} < y_i^h, \text{ for } i = 1, 2, j = 3 - i. \quad (30)$$

Next, we characterize the equilibrium for the firms. Following Tang and Kouvelis (2011), we first derive the unconstrained equilibrium in the production and forward games by ignoring condition (30). Using Equations (28) and (29), we obtain the equilibrium forward quantity and target output for each firm as follows:

$$f^* = \frac{aA - 144bB}{2C}, \quad q^* = \frac{aD - 16bE}{(1 - \delta)C}, \quad (31)$$

where $A = (1 - \delta)(425 + 517\rho + 187\rho^2 + 23\rho^3)$, $B = 3 + 10\rho + 3\rho^2$, $C = (1 - \delta)(1121 + 1297\rho + 427\rho^2 + 35\rho^3)$, $D = (1 - \delta)(947 + 1015\rho + 313\rho^2 + 29\rho^3)$, and $E = 287 + 210\rho + 31\rho^2$.

Similar to the baseline model, to avoid triviality we assume the market size is large enough so that the equilibrium target output is positive. This amounts to the condition $a > \frac{16bE}{D}$. As discussed earlier, condition (30) is required for Pattern II. Thus, substituting the equilibrium forward quantity and target output into the condition, we obtain the equilibrium condition for Pattern II: $\frac{16b((3+\delta)E-9(1-\delta)B)}{(3+\delta)D-(1-\delta)(A+2C)} < a < \frac{48b(E-3B)}{3D-A-2C}$. Note that $\frac{16b((3+\delta)E-9(1-\delta)B)}{(3+\delta)D-(1-\delta)(A+2C)} > \frac{16bE}{D}$.

6.2.1 | Effect of yield uncertainty

Having characterized the equilibrium for the firms, we now analyze the impacts of yield uncertainty on the equilibrium outcomes. Since analytical analysis is not amenable given the complex expressions of equilibrium outcomes, we will resort to a numerical approach. To save space and highlight the impact of withholding behavior, we will only present the comparison results of sensitivity analyses between our baseline model and the withholding model in Table 2. The detailed numerical results are illustrated using figures in Supporting Information Appendix F.

From Table 2, we find that the following interesting results in the baseline model still exist in the withholding model: (1) As yield risk increases, firms sell less in the forward market,

TABLE 2 Effects of yield uncertainty in the baseline and withholding models

	Baseline model (no withholding)		Withholding model	
	δ	ρ	δ	ρ
q^*	$\downarrow$	$\downarrow$	$\searrow$	$\downarrow$ when δ is small; $\searrow$ when δ is medium; $\uparrow$ when δ is large
f^*	$\downarrow$	$\uparrow$	$\downarrow$	$\uparrow$ when δ is low; $\searrow$ when δ is medium; $\downarrow$ when δ is large
R	$\downarrow$ when $\rho = 0$; $\searrow$ when $0 < \rho < 1/2$; $\uparrow$ when $\rho \geq 1/2$	$\uparrow$	$\searrow$	$\uparrow$ when δ is low; $\searrow$ when δ is medium; $\downarrow$ when δ is high
Π^*	$\searrow$ when $\rho < 1/2$; $\downarrow$ when $\rho \geq 1/2$	$\downarrow$	$\searrow$	$\uparrow$ when δ is low; $\searrow$ when δ is medium; $\downarrow$ when δ is high
$E[p^*]$	$\uparrow$	$\uparrow$	$\uparrow$	$\uparrow$
$\text{Var}[p^*]$ and $\text{cv}[p^*]$	$\searrow$	$\uparrow$	$\uparrow$	$\uparrow$ when δ is low; $\searrow$ when δ is medium; $\downarrow$ when δ is high

and the forward ratio can decrease in the yield risk; (2) higher yield risk may benefit the firms; and (3) the expected spot price increases in the yield risk and correlation.

However, there are some contrasting sensitivity results between the baseline and withholding models as shown in Table 2. In fact, all these contrasting results can be explained by a new driving force due to the firms' withholding behavior: *the flexibility of withholding in spot transaction and the lowered effective yield in the spot market for a given target output*. The value of withholding flexibility in spot transaction is higher as yield risk or correlation (δ or ρ) increases. According to the two-point yield distribution, as δ increases, y_i^l (equal to $\frac{1}{2} - \frac{\delta}{2}$) decreases but y_i^h (equal to $\frac{1}{2} + \frac{\delta}{2}$) increases. However, each firm will withhold a portion of its products when realizing a high yield. As δ increases, the withholding quantity is larger, which implies a higher flexibility value. Similarly, with a larger ρ , it is more likely that both firms realize a low yield, or both realize a high yield but in this case they withhold more products than in the case with asymmetric yield realizations. Therefore, a larger ρ also leads to a greater flexibility value.

Equipped with the understanding of the new driving force, we are ready to explain in detail the contrasting results between the baseline and withholding models in Table 2.

Target output, forward quantity, and forward ratio

Table 2 shows that unlike the baseline model, q^* may increase in δ when δ is low in the withholding model. As claimed above, the firms enjoy the flexibility of withholding some products when the yield turns out to be high. An increase in yield risk enhances this operational flexibility and hence the marginal revenue of output, which can lead to higher target output. However, as δ becomes very large, the yield is either very low or very high (in which case withholding occurs), and the effective yield of output becomes extremely low. There-

fore, for the range of high δ values, the marginal value of the target output decreases sharply in δ , and so does q .

The effects of yield correlation on the target output and forward quantity are nonmonotone in the withholding model. In the baseline model, we have shown that a higher correlation leads to a lower marginal revenue of output and hence each firm reduces its target output when ρ increases. This driving force still exists in the withholding model. In addition, with the withholding behavior, a larger ρ leads to a greater flexibility value of withholding, and hence the firms have incentives to increase their target outputs. This effect is more pronounced with a larger δ , and hence when δ is large enough, the firms produce more for a larger ρ . But when δ is intermediate, the joint effect of these forces leads to the nonmonotone effect of ρ on the target output.

Since a larger ρ leads to a greater flexibility value in a spot transaction, a spot transaction becomes more attractive relative to the forward transaction. As a result, the firms' forward quantities may decrease in ρ as well.

The forward ratio is the ratio of forward quantity to the target output. In the withholding model, a spot transaction is more attractive for a greater value of ρ . As a result, compared to the baseline model, the decreasing trend of the forward ratio R in ρ becomes possible in the withholding model.

In terms of the impact of δ on R , R always first decreases and then increases in δ in the withholding model. The decreasing part is due to a more attractive spot transaction as δ increases in the withholding model. The increasing part is mainly due to the fact that target output q decreases sharply in δ for a high range of δ values in the withholding model.

Profit

From Table 2, in the withholding model, the impact of yield risk and correlation on the firm profit is opposite to their

impact on the target output. This is because some products may be withheld in the event of high yield realizations, which drives up the *effective* unit production cost. As a result, in equilibrium the firms may benefit from less production but be hurt by more production considering the prisoner's dilemma issue.

Spot price

Table 2 shows that yield uncertainty has the same effect on the expected spot price in both the baseline and withholding models. We now explain the contrasting results on spot price volatility. Since $\text{Var}[p^*] = \text{Var}[a - f_1^* - f_2^* - s_1^* - s_2^*] = \text{Var}[s_1^* + s_2^*]$, in the baseline model the spot price volatility is determined by the volatility of traded quantity in the spot market. A higher yield risk directly leads to higher volatility of quantity traded in the spot market, but indirectly may decrease the target output and the variance of aggregate yield. In the withholding model, the spot price volatility is not affected by the target output as much as in the baseline model because of the withholding behavior, and as a result the decreasing trend of spot price volatility in yield risk disappears and the spot price volatility always increases in yield risk.

The impact of ρ presents more patterns. On the one hand, a higher yield correlation leads the aggregate output to be more volatile due to the lack of risk pooling effect, which increases spot price volatility. On the other hand, the withholding behavior leads to a lower effective yield in the spot market, and the effect of withholding is stronger with a higher ρ . Hence, the spot price volatility may decrease in ρ in the withholding model.

6.2.2 | Role of a forward market

In the baseline model, the existence of a forward market decreases firms' equilibrium profits. In the withholding model, by calculating and comparing the equilibrium results with and without a forward market, we find that a forward market again leads to smaller equilibrium profits for the firms. The underlying driving force is similar to that in the baseline model. That is, the first-mover advantage (i.e., q_i increases in f_i), which is used to analyze the effect of a forward market on firms' profits in the nonwithholding model, carries over in the withholding model. Hence, firms enjoy the first-mover advantage by selling forward, which leads to higher target outputs and lower spot price and end up with a lower profit in equilibrium. Furthermore, we find that a forward market again destabilizes the spot price, which is driven by the same force as in the baseline model. Adopting the notation used in the nonwithholding model, we formalize these results in Proposition 10 below.

Proposition 10 (Role of a forward market in the withholding model). $\Pi^* < \mathbb{Z}^*$ and $\text{Var}[p^*] > \text{Var}[P^*]$.

7 | CONCLUSIONS AND FUTURE RESEARCH

This paper is primarily motivated by the agricultural sector in which crop yield is intrinsically uncertain, and firms can trade in both forward and spot markets. We have studied the impacts of yield uncertainty on the market structure and spot price characteristics. We also examine how yield uncertainty mediates the role of a forward market in determining the equilibrium outcomes.

Some interesting results have emerged from this research. The impact of yield risk on firms' profits depends on the correlation between firms' yields. Specifically, an increase in yield risk may benefit firms in equilibrium when firms' yields are not too systemic, and the initial yield risk is low. The intuition is that having a slightly higher yield risk may mitigate the competition between firms. Conventional wisdom may suggest that firms should engage in more forward transactions in a more uncertain environment since a forward market allows firms to lock the price at a certain level. Our analysis shows that a higher yield risk leads to a higher forward ratio only when either the yield risk or yield correlation is high. For low yield correlation, the forward ratio can be lower for a higher yield risk, since in this way, firms can take advantage of the chance that the spot price may be high when the other firm's realized yield is low.

We also show that the impact of yield risk on spot price volatility is nonmonotone. Price volatility in our model is driven by two factors: each firm's target production quantity and the yield variance. Since the target quantity decreases in the variance of yield, the combined effect of the two factors explains the nonmonotone impact of yield risk on spot price volatility. In addition, the spot price is more volatile when the yield correlation is higher under the two-firm setting. However, as the number of firms becomes large, a higher yield correlation may lead to a lower spot price volatility. This is because spot price volatility is positively correlated with the aggregate output volatility. A higher yield correlation on the one hand directly increases aggregate output volatility due to the lack of pooling, but on the other hand decreases firms' output quantities and hence indirectly decreases output volatility. The latter effect can be strong enough when there are many firms in the market, which leads to a lower aggregate output volatility and hence a lower spot price variability.

We also analyze how the existence of a forward market affects the equilibrium outcome and how this effect is mediated by yield uncertainty. Interestingly, the spot price becomes more volatile in the presence of a forward market, a result counter to our intuitive understanding. This is because firms produce more in the presence of a forward market, resulting in a spot price of lower expected value but higher volatility. In addition, we find that a higher yield risk mitigates the impact of a forward market on all aspects of the equilibrium. For example, a higher yield risk mitigates the

prisoner's dilemma caused by the existence of a forward market. In contrast, a higher yield correlation enhances the role of a forward market and exacerbates the prisoner's dilemma.

Finally, we have extended the study to the setting where the firms may withhold products in the spot market. The results derived in the baseline model show that higher yield risk leads firms to sell less in the forward market and counter-intuitively may benefit firms carry over to the withholding model. Furthermore, in both baseline and withholding models, a forward market destabilizes spot prices and decreases firms' profits. On the other hand, unlike the baseline model, the effects of yield correlation can be nonmonotone in the withholding model. This is due to the fact that the withholding option provides the firms with additional leverage to deal with the high yield scenarios and a higher yield correlation decreases each firm's effective yield, implying a greater value of withholding flexibility.

This research can be extended in several directions. First, to focus on the interplay of yield uncertainty and forward market, our model considers risk-neutral firms. It is interesting to examine whether risk attitude would alter the results obtained in this paper. Second, this paper considers stochastic proportional yield and the result that the forward market destabilizes spot prices largely depends on this assumption in a competitive setting. How the results will change when considering additive yield uncertainty deserves more future research. Third, several hypotheses can be developed based on the analytical results about the impacts of yield uncertainty and forward market on market structure and spot price volatility. We expect to see future empirical studies that test those hypotheses. Lastly, the current model studies a hybrid market with forward and spot transactions. In some industries, firms may trade in an options market (e.g., Anderson et al., 2017; Wu & Kleindorfer, 2005). It would be interesting to study whether some of our findings carry over to markets with spot and option transactions.

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ENDNOTES

¹The firms engage in a Stackelberg game by first making their forward quantity decisions and then making target output decisions. Therefore, the benefit of trading in the forward market is referred to as the first-mover advantage.

²For tractability, we consider symmetric yield distributions. We have numerically checked that our key results remain in the following two extensions: (1) a two-firm setting where the firm yields have different means but the same variance and (2) a three-firm setting where the yields share the same distribution (with the same mean and variance) but the cor-

relation coefficients may be different. The detailed derivation and analysis are provided in Supporting Information Appendix C.

³In this paper, we consider forward markets instead of futures markets for two reasons: First, futures contracts involve more complex technical details. Futures contracts are marking to market, that is, futures price changes result in daily gains or losses, which are credited to or subtracted from the margin account. Second, forward contracting historically has been the principal risk management tool of crop farmers in the United States (Davis et al., 2005; Etienne et al., 2017; Goodwin & Schroeder, 1994). For some agricultural products, there are no relevant futures markets, but they can be traded in forward markets.

⁴We have also analyzed the case of convex production cost and obtained the same sensitivity results about the firms' equilibrium profits, forward ratios, and spot price volatility as those in the case of linear production cost.

⁵The GRDC is one of the world's leading grains research organizations, responsible for planning, investing in and overseeing Research, Development & Extension Plan to deliver improvements in production, sustainability, and profitability across the Australian grains industry. The GRDC is accountable to the Australian Parliament through the Minister for Agriculture and Water Resources. (See <https://grdc.com.au/about/who-we-are> for details.)

⁶Firm i 's forward price for actual users is equal to that for speculators, and hence firm i 's revenue from forward transaction does not depend on α_i .

⁷The same result has been established for our benchmark in Subsection 3.1.

⁸To show the existence of first-mover advantage, we have analytically studied an extension of our baseline model in which only one firm has access to the forward market and the rest of the model setup remains unchanged. The detailed derivation and analysis are provided in Supporting Information Appendix C. We find that the firm with access to the forward market enjoys the first-mover advantage by targeting a larger output, but the other firm decreases its target output consequently.

⁹To show the existence of the nonmonotone result, we have examined two yield distributions over $[0,1]$: a two-point distribution and a triangular distribution. For the detailed derivation and analysis, refer to Supporting Information Appendix B. We find that the nonmonotone impact of δ does exist for certain parameter settings. For example, consider the two-point yield distribution: $Pr(Y_i = a) = p$, $Pr(Y_i = b) = 1 - p$, $i = 1, 2$, where $0 \leq p \leq 1$ and $0 \leq a \leq b \leq 1$. The upper bound of δ is $\sqrt{(1-\mu)/\mu}$, which decreases in μ . Hence, δ has a nonmonotone effect if $\sqrt{(1-\mu)/\mu}$ is higher than the pattern switching threshold of δ , which corresponds to sufficiently small μ .

¹⁰For the detailed real-world empirical evidence, refer to Supporting Information Appendix D.

¹¹Note that in the asymmetric model where only one firm can trade in the forward market, the nonmonotone effect of yield risk on the firms' profits disappears. The reason is that, in the asymmetric case, there is no competition between the two firms in the forward market, and hence the prisoner's dilemma is absent.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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